



2024

HSC Assessment Task 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour
- Write using non erasable black pen.
- Calculators approved by NESA may be used.
- Answer all questions in the booklet provided.
- In Questions 6-8, show relevant mathematical reasoning and/or calculations.

Total Marks:
31

Section I – 5 marks (pages 2-3)

- Attempt Questions 1-5
- Allow about 7 minutes for this section

Section II - 26 marks (pages 4-6)

- Attempt Questions 6 - 8
- Allow about 53 minutes for this section
- Marks may be deducted for careless or badly arranged work

Section I

5 marks

Attempt Questions 1 – 5

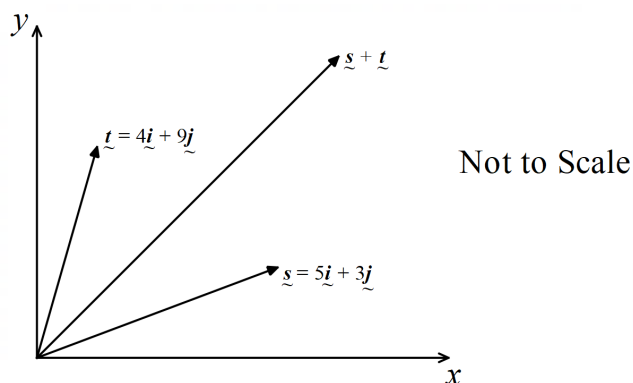
Allow about 7 minutes for this section

Use the multiple-choice answer page in the writing booklet for Questions 1-5.

- 1 Given the points $A(-3, -5)$ and $B(0, 7)$, which of the following represents the displacement vector $\overrightarrow{BA}$?

- (A)
- (B)
- (C) $3\mathbf{i} - 2\mathbf{j}$
- (D) $-3\mathbf{i} + 2\mathbf{j}$

- 2 The diagram below shows the vectors $\mathbf{s}$, $\mathbf{t}$ and $\mathbf{s} + \mathbf{t}$.



What is the value of $|\mathbf{s} + \mathbf{t}|$?

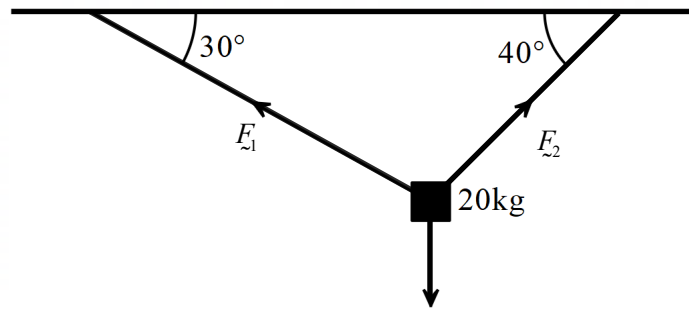
- (A) 12
- (B) 15
- (C) 16
- (D) 20

- 3 Given $\mathbf{a}$ is a vector such that $\mathbf{a} = x\mathbf{i} + y\mathbf{j}$ where x and y are real scalars.

$\mathbf{b}$ is a vector on the line $x = 0$, which of the following is the projection of $\mathbf{a}$ onto $\mathbf{b}$?

- (A) $x\mathbf{i}$
- (B) $x\mathbf{j}$
- (C) $y\mathbf{i}$
- (D) $y\mathbf{j}$

- 4 A particle of mass 20kg is suspended by two strings attached to two points in the same horizontal plane. The two strings make angles of 30° and 40° to the horizontal, as shown below. The tension in the strings are F_1 and F_2 , measured in newtons. (You may assume $g = 9.8\text{ m/s}^2$)



Which of the following is closest to the magnitude of the tension of the force in each string in newtons.

- (A) $|F_1| = 160\text{ N}$ and $|F_2| = 181\text{ N}$
(B) $|F_1| = 181\text{ N}$ and $|F_2| = 160\text{ N}$
(C) $|F_1| = 84\text{ N}$ and $|F_2| = 112\text{ N}$
(D) $|F_1| = 112\text{ N}$ and $|F_2| = 84\text{ N}$
-
- 5 Mathematical induction is used to prove that $n(2n - 1)(2n + 1)$ is divisible by 3 for all integers $n \geq 1$. Which of the following is the inductive statement that needs to be proved?
- (A) $3k(2k - 1)(2k + 1)$ is divisible by 3.
(B) $(k + 1)(2k)(2k + 2)$ is divisible by 3.
(C) $(k + 1)(2k + 1)(2k + 2)$ is divisible by 3.
(D) $(k + 1)(2k + 1)(2k + 3)$ is divisible by 3.
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Section II**Marks****26 marks****Attempt Questions 6-8****Allow about 53 minutes for this section****Answer each question in the appropriate section of the writing booklet.****Extra writing paper is available.****Your responses should include relevant mathematical reasoning and/or calculations.****Question 6 (10 marks) Use the Question 6 section of the writing booklet.**a) Let $\underline{a} = 3\underline{i} + \underline{j}$ and $\underline{b} = 2\underline{i} - \underline{j}$.

- | | | |
|-------|--|---|
| (i) | Calculate the magnitude of $\underline{a}$. | 1 |
| (ii) | Find $\hat{a}$. | 1 |
| (iii) | Find the vector projection of $\underline{b}$ onto $\underline{a}$. | 1 |
| (iv) | Write a possible vector that is perpendicular to $\underline{b}$. | 1 |

b) Given the vectors $\overrightarrow{OA} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, $\overrightarrow{OB} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\overrightarrow{OC} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$,
calculate the size of $\angle ABC$ correct to the nearest degree. 3

c) Use mathematical induction to prove that $23^n - 1$ is divisible by 11 for all positive integers n . 3

End of Question 6

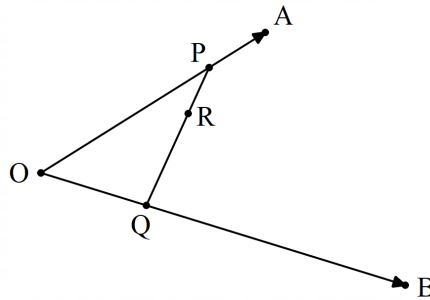
Question 7 (8 marks) Use the Question 7 section of the writing booklet.

- a) Three forces act on an object of mass 2 kg. These forces, measured in newtons, are represented by the vectors 2

$$\begin{bmatrix} 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ -10 \end{bmatrix} \text{ and } \begin{bmatrix} -2 \\ -20 \end{bmatrix}.$$

Calculate the magnitude of the acceleration of the object.

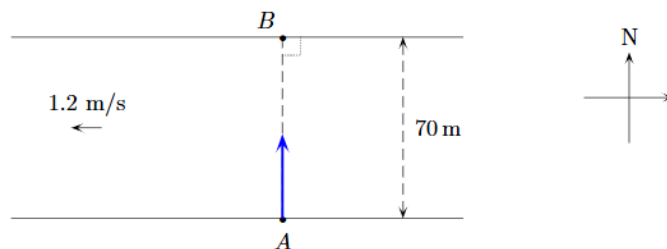
- b) In the diagram, $\overrightarrow{OP} = (1-\lambda)\overrightarrow{OA}$, $\overrightarrow{OQ} = \lambda\overrightarrow{OB}$ and $\overrightarrow{PR} = \lambda\overrightarrow{PQ}$, for some real number λ where $0 \leq \lambda \leq 1$.



- (i) Show that $\overrightarrow{OR} = (1-\lambda)^2\overrightarrow{OA} + \lambda^2\overrightarrow{OB}$ where $0 \leq \lambda \leq 1$. 1

- (ii) Find the Cartesian equation of the curve described by $\overrightarrow{OR}$ when $\overrightarrow{OA} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\overrightarrow{OB} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$. 2

- c) The banks of the river are parallel and 70m apart. Point B is directly opposite point A on the other side of the river, as shown below.



In still water, a yacht can travel at a speed of 1.8 m/s.
The river is flowing downstream at 1.2 m/s, and there is a wind that blows at a constant speed of 0.5 m/s in the north-westerly direction.

- (i) Draw a vector diagram labelling all forces. 1
- (ii) On which bearing should the yacht sail in order for it to sail a direct path from point A to point B? Give your answer correct to the nearest degree. 2

End of Question 7

Question 8 (8 marks) Use the Question 8 section of the writing booklet.

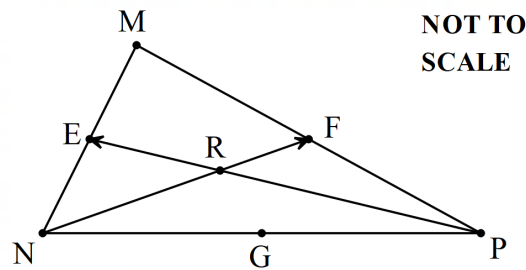
- a) (i) By considering the sum of the terms of an arithmetic series, show that 1

$$(1 + 2 + \dots + n)^2 = \frac{1}{4}n^2(n+1)^2$$

- (ii) By using the Principle of Mathematical induction, prove that 3

$$1^3 + 2^3 + \dots + n^3 = (1 + 2 + \dots + n)^2, \text{ for all integers where } n \geq 1$$

- b) In the triangle MNP shown in the diagram, the points E, F, G are the midpoints of MN , MP and NP respectively. 4

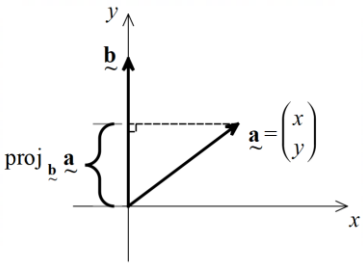
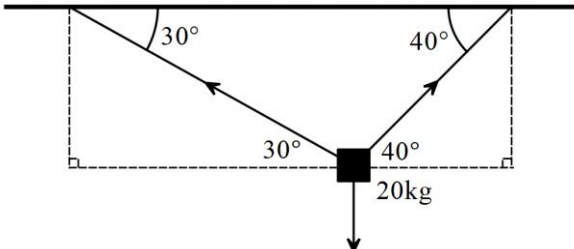


Let $\overrightarrow{MN} = \underline{q}$, $\overrightarrow{MP} = \underline{p}$, $\overrightarrow{NR} = \lambda \overrightarrow{NF}$, and $\overrightarrow{PR} = \mu \overrightarrow{PE}$, where λ and μ are real numbers between 0 and 1.

By finding two expressions for $\overrightarrow{MR}$, show that the medians $\overrightarrow{PE}$, $\overrightarrow{NF}$ and $\overrightarrow{MG}$ are concurrent and that R divides each median in the ratio of 2:1.

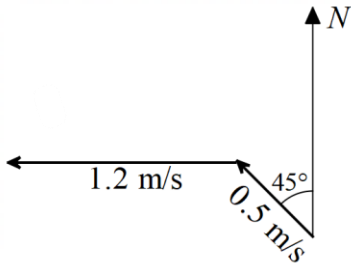
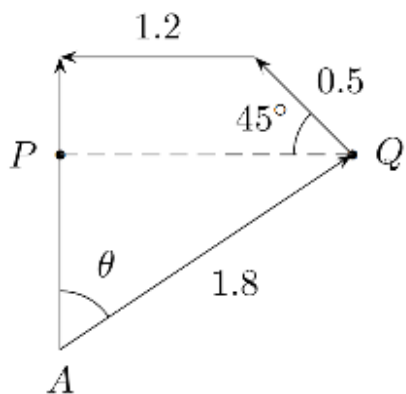
End of Exam

Marking Guideline
Year 12 - Extension 1 - Task 1 2024

Qn	Suggested solutions	Marking Guidelines
1	$\overrightarrow{BA} = \begin{bmatrix} -3 \\ -5 \end{bmatrix} - \begin{bmatrix} 0 \\ 7 \end{bmatrix} = \begin{bmatrix} -3 \\ -12 \end{bmatrix}$	B
2	$ \vec{s} + \vec{t} = 4\vec{i} + 9\vec{j} + 5\vec{i} + 3\vec{j} $ $= 9\vec{i} + 12\vec{j} $ $= \sqrt{9^2 + 12^2}$ $= 15$	B
3		D
4	 Vector sum in the $\vec{i}$ direction: $F_2 \cos 40^\circ - F_1 \cos 30^\circ = 0$ $F_2 = \frac{\cos 30^\circ}{\cos 40^\circ} F_1 \quad (1)$ Vector sum in the $\vec{j}$ direction: $F_1 \sin 30^\circ - F_2 \sin 40^\circ - 20g = 0 \quad (2)$ Sub (1) $\rightarrow$ (2) $ F_1 \sin 30^\circ - \frac{\cos 30^\circ}{\cos 40^\circ} F_1 \times \sin 40^\circ - 20 \times 9.8 = 0$ $ F_1 = 159.8 \text{ N}$ Sub into (1) $ F_2 = 1.1305 \times 159.8$ $ F_2 = 180.6 \text{ N}$ $\therefore F_1 = 160 \text{ N and } F_2 = 181 \text{ N}$	A
5	$(k+1)(2[k+1]-1)(2[k+1]+1) = (k+1)(2k+1)(2k+3)$	D

6a(i)	$ a = \sqrt{3^2 + 1^2} = \sqrt{10}$	1 mark Correct solution
6a(ii)	$\hat{a} = \frac{\underline{a}}{ \underline{a} }$ $\hat{a} = \frac{3\hat{i} + 1\hat{j}}{\sqrt{3^2 + 1^2}}$ $\hat{a} = \frac{3}{\sqrt{10}}\hat{i} + \frac{1}{\sqrt{10}}\hat{j}$	1 mark Correct solution
6a(iii)	$\text{proj}_{\underline{a}} \underline{b} = \frac{\underline{b} \cdot \underline{a}}{\underline{a} \cdot \underline{a}} \underline{a}$ $= \frac{3 \times 2 + 1 \times -1}{3^2 + 1^2} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ $= \begin{bmatrix} 1.5 \\ 0.5 \end{bmatrix}$	1 mark Correct solution
6a(iv)	$\begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = 0$ $2x_1 - y_1 = 0$ $2x_1 = y_1$ must be in the form $\begin{bmatrix} a \\ 2a \end{bmatrix}$, where a is a real number eg. $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	1 mark Correct solution
6b	$\overrightarrow{BA} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \quad \overrightarrow{BC} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $\cos \angle ABC = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{ \overrightarrow{BA} \overrightarrow{BC} }$ $\cos \angle ABC = \frac{2 \times -2 + 2 \times 1}{\sqrt{(-2)^2 + 2^2} \times \sqrt{2^2 + 1^2}}$ $\cos \angle ABC = \frac{-2}{\sqrt{8} \times \sqrt{5}}$ $\angle ABC = 108.43...^\circ$ $\angle ABC = 108^\circ$ (nearest degree)	3 marks - Correct solution. 2 marks - Calculates the dot product of $\overrightarrow{BA}$ and $\overrightarrow{BC}$. 1 mark - Finds $\overrightarrow{BA}$ or $\overrightarrow{BC}$.

6c	Test for $n = 1$ $\text{LHS} = 23^1 - 1$ $= 11 \times 2$ $\therefore \text{True for } n = 1$ Assume true for $n = k$ ie. $23^k - 1 = 11P$, where P is an integer. Prove true for $n = k + 1$ AIM : $23^{k+1} - 1 = 11Q$, where Q is an integer. Proof: $\text{LHS} = 23^{k+1} - 1$ $= 23^k \times 23 - 1$ $= (11P + 1) \times 23 - 1 \quad \text{if the assumption is true}$ $= 23 \times 11P + 22$ $= 11(23P + 2)$ $= 11Q \quad \text{since } (23P + 2) \text{ is an integer}$ $= \text{RHS}$ $\therefore$ true for $n = k + 1$, if it is true for $n = k$. $\therefore$ result is true by Mathematical Induction for all integers $n \geq 1$, since it is true for $n = 1$.	3 marks - Correct induction. 2 marks - Uses the assumption to prove the required result. 1 mark - Correct base case.
7a)	Resultant vector $= \begin{bmatrix} 5 \\ 6 \end{bmatrix} + \begin{bmatrix} 7 \\ -10 \end{bmatrix} + \begin{bmatrix} -2 \\ -20 \end{bmatrix} = \begin{bmatrix} 10 \\ -24 \end{bmatrix}$ magnitude $= \sqrt{10^2 + (-24)^2}$ $= 26$ $F = ma$ $26 = 2a$ $a = 13 \text{ms}^{-2}$	2 marks - Correct solution 1 mark - Find the resultant vector
7b(i)	$\overrightarrow{OR} = \overrightarrow{OP} + \overrightarrow{PR}$ $\overrightarrow{OR} = (1 - \lambda)\overrightarrow{OA} + \lambda\overrightarrow{PQ} \quad \text{---(1)}$ $\overrightarrow{PQ} = \lambda\overrightarrow{OB} - (1 - \lambda)\overrightarrow{OA}$ sub $\overrightarrow{PQ}$ into (1) $\overrightarrow{OR} = (1 - \lambda)\overrightarrow{OA} + \lambda[\lambda\overrightarrow{OB} - (1 - \lambda)\overrightarrow{OA}]$ $\overrightarrow{OR} = (1 - 2\lambda + \lambda^2)\overrightarrow{OA} + \lambda^2\overrightarrow{OB}$	1 marks Correct solution

7b(ii)	$\overrightarrow{OR} = (1-\lambda)^2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \lambda^2 \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ $= \begin{bmatrix} 1-2\lambda \\ 2-4\lambda+5\lambda^2 \end{bmatrix}$ $x = 1-2\lambda \quad \text{and} \quad y = 2-4\lambda+5\lambda^2$ $\lambda = \frac{1-x}{2}$ sub λ into y $y = 2-4\left(\frac{1-x}{2}\right) + 5\left(\frac{1-x}{2}\right)^2$ $y = \frac{5x^2}{4} - \frac{x}{2} + \frac{5}{4}$	2 marks - Correct solution 1 mark - Finds the correct parametric equations
7c(i)		1 mark Correct solution
7c(ii)	Forces parallel to the river bank ie. $\overrightarrow{PQ}$:  $1.8 \sin \theta - 0.5 \cos 45^\circ - 1.2 = 0$ In $\triangle APQ$, $\sin \theta = \frac{\frac{6}{5} + \frac{1}{2\sqrt{2}}}{1.8}$ $\theta = 59.6647..^\circ$ $\therefore$ The bearing that the fish should head to is $060^\circ \text{T (to the nearest degree)}$ $1.8 \cos \theta + 0.5 \sin 45^\circ$	2 marks - Correct solution 1 mark - Finds $\overrightarrow{PQ}$

8a(i)	$S_n = \frac{n}{2}(a+l)$ $1+2+\dots+n = \frac{n}{2}(1+n)$ square both sides $(1+2+\dots+n)^2 = \frac{n^2}{4}(1+n)^2$	1 mark Correct solution
8a(ii)	Test for $n = 1$ $\text{LHS} = 1^3 \quad \text{RHS} = 1^2$ $= 1 \quad = 1$ $\therefore$ True for $n = 1$ Assume true for $n = k$ ie. $1^3 + 2^3 + \dots + k^3 = (1+2+\dots+k)^2$ Prove true for $n = k+1$ <i>AIM</i> : $1^3 + 2^3 + \dots + k^3 + (k+1)^3 = (1+2+\dots+k+[k+1])^2$ <i>Proof</i>: $\text{LHS} = 1^3 + 2^3 + \dots + k^3 + (k+1)^3$ $= (1+2+\dots+k)^2 + (k+1)^3 \quad \text{if the assumption is true}$ $= \frac{k^2}{4}(k+1)^2 + (k+1)^3 \quad \text{using part (i)}$ $= \frac{(k+1)^2}{4} [k^2 + 4(k+1)]$ $= \frac{(k+1)^2}{4} [k+2]^2$ $\text{RHS} = (1+2+\dots+k+[k+1])^2$ $= \left[\frac{(k+1)}{2} (1+k+1) \right]^2$ $= \frac{(k+1)^2}{4} (k+2)^2$ $= \text{LHS}$ $\therefore$ true for $n = k+1$, if it is true for $n = k$. $\therefore$ result is true by Mathematical Induction for all integers $n \geq 1$, since it is true for $n = 1$.	3 marks - Correct induction. 2 marks - Uses the assumption to prove the required result. 1 mark - Correct base case, showing correct substitution. Note: Marks were deducted if proving for $1^3 + 2^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$ without linking to the question.

8b(iii)

Here are 4 ways we can obtain the vector $\overrightarrow{MR}$

$$\overrightarrow{MR} = \overrightarrow{MN} + \overrightarrow{NR}$$

$$\overrightarrow{MR} = \underline{n} + \lambda \overrightarrow{NF} \quad - (1)$$

$$\overrightarrow{MR} = \overrightarrow{MP} + \overrightarrow{PR}$$

$$\overrightarrow{MR} = \underline{p} + \mu \overrightarrow{PE} \quad - (2)$$

$$\overrightarrow{MR} = \overrightarrow{ME} + \overrightarrow{ER}$$

$$\overrightarrow{MR} = \frac{1}{2} \underline{n} - (1 - \mu) \overrightarrow{PE} \quad - (3)$$

$$\overrightarrow{MR} = \overrightarrow{MF} + \overrightarrow{FR}$$

$$\overrightarrow{MR} = \frac{1}{2} \underline{p} - (1 - \lambda) \overrightarrow{NF} \quad - (4)$$

$$\overrightarrow{NF} = -\underline{n} + \frac{1}{2} \underline{p}$$

$$\overrightarrow{PE} = -\underline{p} + \frac{1}{2} \underline{n}$$

Sub $\overrightarrow{NF}$ into (1) or (4)Sub $\overrightarrow{NF}$ into (1):

$$\overrightarrow{MR} = \underline{n} + \lambda \left(-\underline{n} + \frac{1}{2} \underline{p} \right)$$

$$\overrightarrow{MR} = (1 - \lambda) \underline{n} + \frac{1}{2} \lambda \underline{p} \quad - (5)$$

Sub $\overrightarrow{NF}$ into (4):

$$\overrightarrow{MR} = \frac{1}{2} \underline{p} - (1 - \lambda) \left(-\underline{n} + \frac{1}{2} \underline{p} \right)$$

$$\overrightarrow{MR} = \frac{1}{2} \underline{p} + \underline{n} - \frac{1}{2} \underline{p} - \lambda \underline{n} + \frac{1}{2} \lambda \underline{p}$$

$$\overrightarrow{MR} = (1 - \lambda) \underline{n} + \frac{1}{2} \lambda \underline{p}$$

Sub $\overrightarrow{PE}$ into (2) or (3)Sub $\overrightarrow{PE}$ into (2):

$$\overrightarrow{MR} = \underline{p} + \mu \left(-\underline{p} + \frac{1}{2} \underline{n} \right)$$

$$\overrightarrow{MR} = (1 - \mu) \underline{p} + \frac{1}{2} \mu \underline{n} \quad - (6)$$

Sub $\overrightarrow{PE}$ into (3):

$$\overrightarrow{MR} = \frac{1}{2} \underline{n} - (1 - \mu) \left(-\underline{p} + \frac{1}{2} \underline{n} \right)$$

$$\overrightarrow{MR} = \frac{1}{2} \underline{n} + \underline{p} - \frac{1}{2} \underline{n} - \mu \underline{p} + \frac{1}{2} \mu \underline{n}$$

$$\overrightarrow{MR} = (1 - \mu) \underline{p} + \frac{1}{2} \mu \underline{n}$$

using (5) and (6) to equate $\underline{p}$ and $\underline{n}$:

$$\frac{1}{2} \lambda = 1 - \mu \quad - (7)$$

$$\frac{1}{2} \mu = 1 - \lambda \quad - (8)$$

4 marks

- Correct solution by showing $\overrightarrow{MR}$ and $\overrightarrow{MG}$ are collinear and proving ratio of 2:1 for all 3 medians.

3 marks

- Solves for $\lambda = \frac{2}{3}$ and $\mu = \frac{2}{3}$

2 marks

- Finds expression of $\overrightarrow{MR}$ in terms of $\underline{n}$, $\underline{p}$, λ and μ only ie.

$$\overrightarrow{MR} = (1 - \mu) \underline{p} + \frac{\mu}{2} \underline{n}$$

$$\overrightarrow{MR} = (1 - \lambda) \underline{n} + \frac{\lambda}{2} \underline{p}$$

1 mark

- Finds one correct expression of $\overrightarrow{MR}$ using $\underline{n}$ and $\underline{p}$ listed below:

$$\overrightarrow{MR} = \underline{n} + \lambda \overrightarrow{NF}$$

$$\overrightarrow{MR} = \underline{p} + \mu \overrightarrow{PE}$$

$$\overrightarrow{MR} = \frac{1}{2} \underline{n} - (1 - \mu) \overrightarrow{PE}$$

$$\overrightarrow{MR} = \frac{1}{2} \underline{p} - (1 - \lambda) \overrightarrow{NF}$$

Sub (7) into (8)

$$\frac{1}{2}\mu = 1 - 2(1 - \mu)$$

$$\mu = \frac{2}{3}$$

Sub y into (7)

$$\lambda = 2\left(1 - \frac{2}{3}\right) = \frac{2}{3}$$

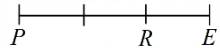
$$\therefore \lambda = \mu = \frac{2}{3}$$

$$\overrightarrow{NR} = \frac{2}{3}\overrightarrow{NF}$$



$$\therefore |\overrightarrow{NR}| : |\overrightarrow{NF}| = 2 : 1$$

$$\overrightarrow{PR} = \frac{2}{3}\overrightarrow{PE}$$



$$|\overrightarrow{PR}| : |\overrightarrow{PE}| = 2 : 1$$

Use (1), (2), (3) or (4) to obtain

$$\overrightarrow{MR} = \frac{1}{3}(p + n)$$

$$\overrightarrow{MG} = \overrightarrow{MN} + \overrightarrow{NG}$$

$$\overrightarrow{MG} = n + \overrightarrow{NG}$$

we know

$$\overrightarrow{NG} = \frac{1}{2}\overrightarrow{NP}$$

$$\overrightarrow{NG} = \frac{1}{2}(p - n)$$

$$\overrightarrow{MG} = n + \frac{1}{2}(p - n)$$

$$\overrightarrow{MG} = \frac{1}{2}(p + n)$$

$$= \frac{3}{2}\overrightarrow{MR}$$

Since $\overrightarrow{MG}$ is a scalar multiple of $\overrightarrow{MR}$
and intersect at the same point M ,
then M, R, G are collinear.

$\begin{aligned} & \left \overrightarrow{MR} \right : \left \overrightarrow{RG} \right \\ &= \left \overrightarrow{MR} \right : \left \overrightarrow{MG} - \overrightarrow{MR} \right \\ &= \frac{1}{3} \left \vec{n} + \vec{p} \right : \frac{1}{2} \left \vec{n} + \vec{p} \right - \frac{1}{3} \left \vec{n} + \vec{p} \right \\ &= \frac{1}{3} \left \vec{n} + \vec{p} \right : \frac{1}{6} \left \vec{n} + \vec{p} \right \\ &= \frac{1}{3} : \frac{1}{6} \\ &= 2 : 1 \\ &\overrightarrow{PE}, \overrightarrow{NF} \text{ and } \overrightarrow{MG} \text{ is concurrent at } M, \text{ in the ratio } 2:1 \end{aligned}$	
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